# Image Caption Generator

Using CNN and LSTM | Deep Learning Internship

PRESENTED BY DARSHINI M S · USN: 4MH22CS026

Maharaja Institute of Technology Mysore

RunShaw Technologies Private Limited

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# What is Image Captioning?

Image captioning sits at the intersection of **Computer Vision** and **Natural Language Processing** — teaching machines to not just see, but describe what they see.

## CNN

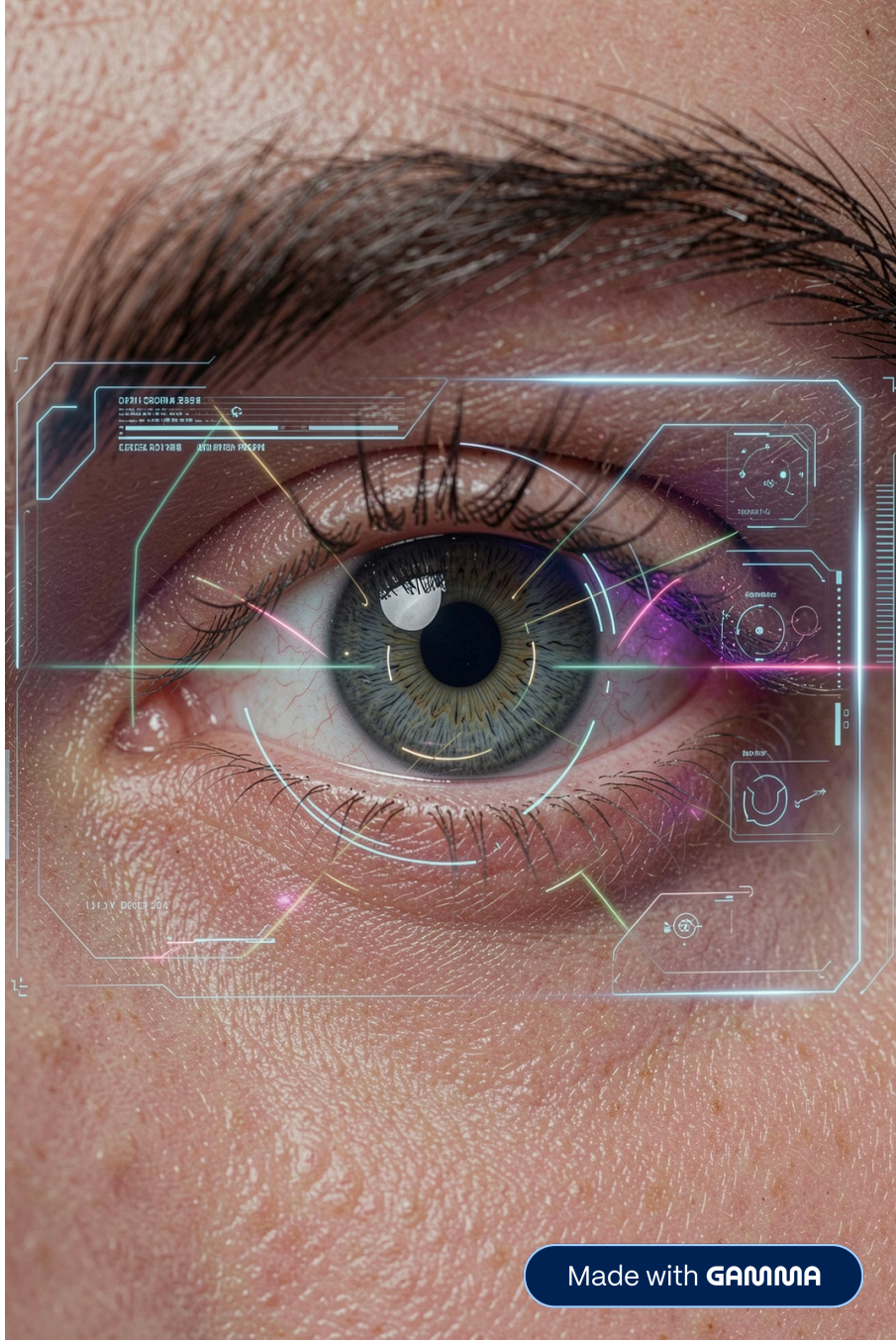
Extracts rich visual features and patterns from input images

## LSTM

Generates fluent, meaningful sentences from extracted features

## Applications

Assistive tech, search engines, and social media automation



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# The Problem We're Solving

## Why Automate Captioning?

Manual image description is **time-consuming, inconsistent, and unscalable**. As visual data grows exponentially, humans can't keep pace.

- Machines struggle to interpret and verbalize visual scenes
- Goal: generate **accurate, human-like captions** automatically
- Must handle diverse, complex, real-world images reliably

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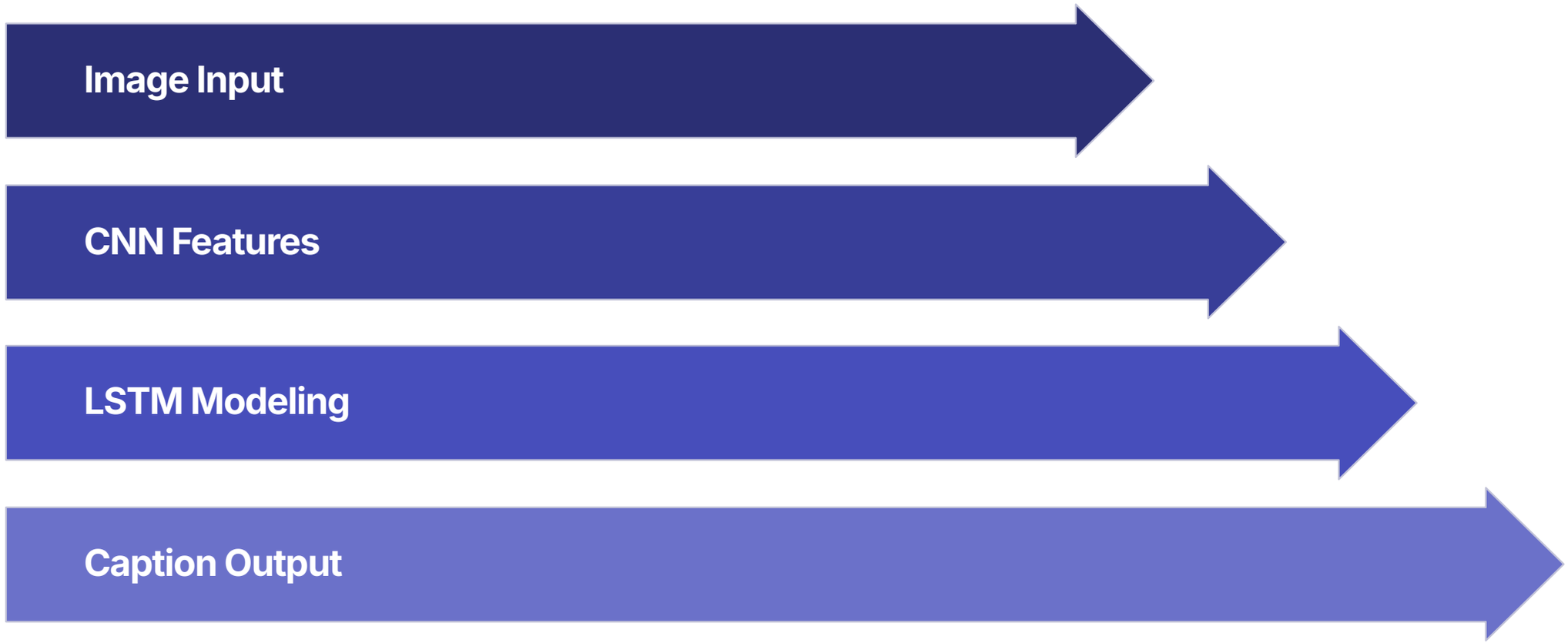
# Methodology

A structured pipeline connects raw images to generated text through two powerful deep learning stages.




The **Flickr8k dataset** provides 8,000 annotated images for training and validation. Captions are tokenized, padded, and mapped to image feature vectors extracted by the pre-trained **VGG16** CNN architecture.

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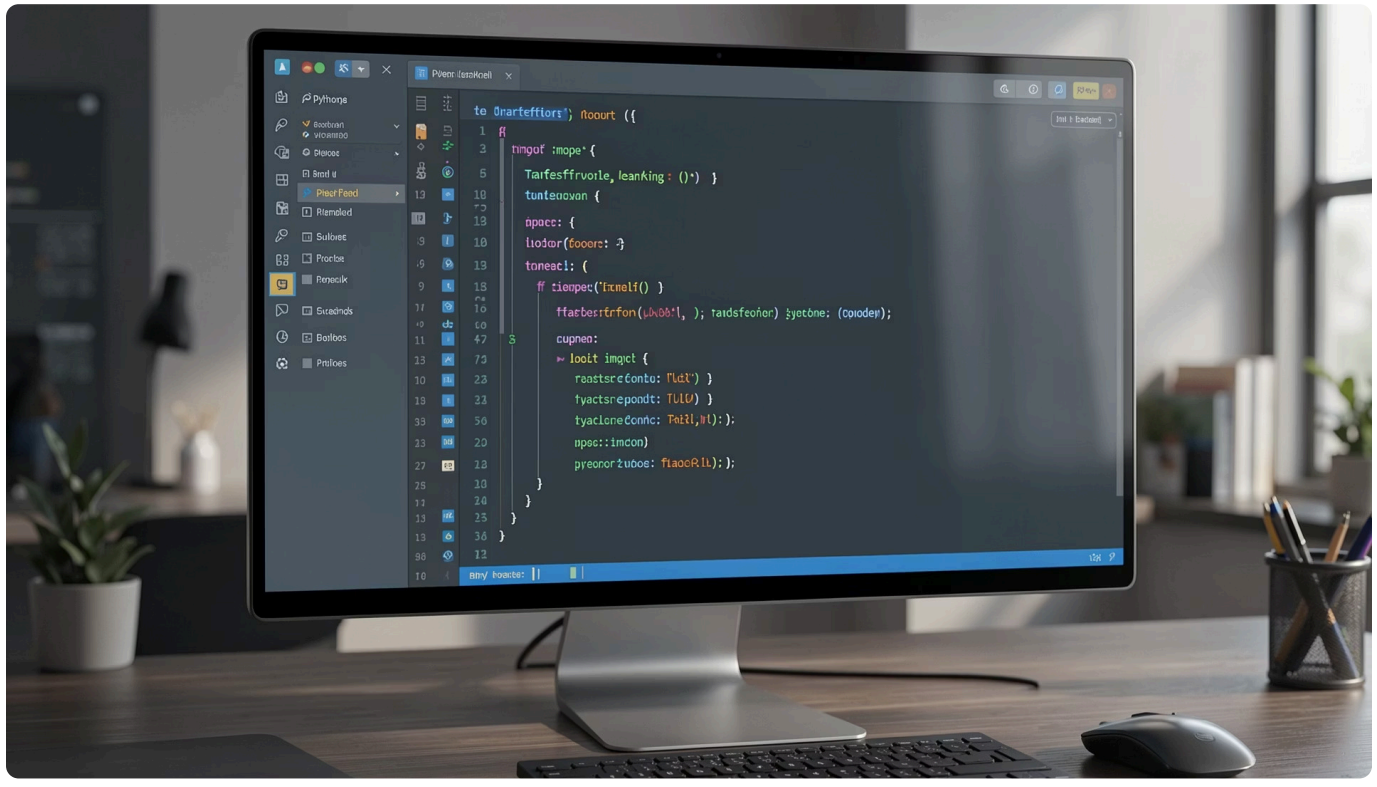
# Implementation

| | |
|---|---|
| 1 | **Feature Extraction**<br>VGG16 pre-trained model extracts 4096-dim feature vectors from each image |
| 2 | **Text Preprocessing**<br>Captions tokenized, vocabulary built, sequences padded to uniform length |
| 3 | **Model Architecture**<br>Image features merged with embedded text input and fed into LSTM layers |
| 4 | **Training**<br>Model trained end-to-end using TensorFlow and Keras with categorical cross-entropy loss |



**Tech Stack:** Python · TensorFlow · Keras · NumPy · Matplotlib

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# Results & Analysis

### Caption Generation

Model successfully generates contextually meaningful captions across diverse test images

### Loss Reduction

Training and validation loss show consistent improvement across epochs, confirming learning

### Consistency

Tested across multiple images; captions are coherent, though complexity may vary

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# Conclusion & Future Scope

## What We Achieved

- Built a working CNN + LSTM image caption generator
- Successfully bridged computer vision and natural language
- Demonstrated meaningful caption generation on real images

## What Comes Next

01

Add **attention mechanisms** for smarter image-text alignment

02

Upgrade to transformer models like **BLIP or GPT-4V**

03

Deploy as a **web or mobile application** for real users

04

Train on **larger datasets** (MS-COCO) for improved accuracy

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# Thank You

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